

Theoretical background of a GRR study.

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For the notations, please see the powerpoint file “Black Belt in Six Sigma – Measurement System Analysis.pptx”.

Distribution of the sample average

Let X be a random variable with mean $E(X)$ and variance $\sigma^2(X)$. Let $\bar{X}$ be the average of a sample of X -values and let n be the sample size. Then $E(\bar{X}) = E(X)$ and $\sigma^2(\bar{X}) = \sigma^2(X)/n$. If n is sufficiently large then $\bar{X}$ follows approximately a normal distribution. The latter result is known as the Central Limit Theorem.

Distribution of the sample range

Let X be normally distributed with mean μ and variance σ^2 . Let R be the range of a sample of X -values and let n be the sample size. Then $E(R) = d_2\sigma$ and $\sigma(R) = d_3\sigma$ where the constants d_2 and d_3 solely depend on the sample size n . For instance for a sample of size $d_2 = 2,326$ and $d_3 = 0,864$.

Average and variance of a χ^2 -distribution

Let X follow a χ^2 -distribution with n degrees of freedom, then $E(X) = n$ and $\sigma^2(X) = 2n$.

Distribution of the average squared range $\bar{R}^2$

Let X be normally distributed with variance σ^2 . Let $\bar{R}$ be the average range based on g samples (aka subgroups) of size m , then $\frac{\nu \bar{R}^2}{d_2^{*2} \sigma^2} \sim \chi^2(\nu)$ where the constants ν and d_2^* solely depend on m and g . For instance for 5 samples of size 3 $d_2^* = 1,73857$ and $\nu = 9,3$. Note that $d_2^* \rightarrow d_2$ for $g \rightarrow \infty$.

We therefore have $E\left(\frac{\nu \bar{R}^2}{d_2^{*2} \sigma^2}\right) = \nu$ and hence $E(\bar{R}^2) = d_2^{*2} \sigma^2$.

We will now use the above to derive the formulas for the average and range method to conduct a GRR study.

Calculation of the $EV(= \sigma)$

The repeatability or equipment variation (EV) can be derived from the average range $\bar{R}$ over all parts and appraisers. Assuming that the EV is independent from the appraiser and part, we have $E(\bar{R}) = E(R) = d_2\sigma = d_2EV$. The best estimate we can make for $E(\bar{R})$ is $\bar{R}$. Hence we estimate EV as $\bar{R}/d_2$ with subgroup size r , as each appraiser measures each part r times.

Calculation of the AV

The reproducibility or appraiser variation (AV) can be derived from the range of the appraiser averages $\bar{X}_{DIFF}$. As there are k appraisers, each producing one average measurement, we have $E(\bar{X}_{DIFF}^2) = d_2^{*2} \sigma^2(\bar{X}_i)$ where d_2^* is based on 1 subgroup of size k .

For the average measurement $\bar{X}_i$ of operator i (all parts n and replications r) we can show (see the powerpoint mentioned above for the proof): $\bar{X}_i = \bar{X}_p + X_{i..} + \bar{X}_r$ with

- $\bar{X}_p$ the average reference value for the selected parts, a constant for the selected parts

- $X_{i..}$ the operator bias
- $\bar{X}_r$ the average measurement error for this appraiser (all parts, all replications)

Consequently we have $\sigma^2(\bar{X}_i) = \sigma^2(X_{i..}) + \sigma^2(\bar{X}_r) = AV^2 + \frac{\sigma^2}{nr} = AV^2 + \frac{EV^2}{nr}$. And also $AV^2 = \frac{\bar{X}_{DIFF}^2}{d_2^{*2}} - \frac{EV^2}{nr}$.

Calculation of the PV

The part variation (PV) can be calculated in the same way as the AV .

Calculation of the TV and the $\%GRR$

The total variation TV is then calculated as $TV^2 = PV^2 + GRR^2 = PV^2 + AV^2 + EV^2$. Finally $\%GRR = GRR/TV$.

When performing the GRR study via ANOVA, the quantities AV , PV and EV can be easily derived from the ANOVA-table as shown in the powerpoint.